

Core Pure Mathematics 1

Notes · Edexcel A Level Further Mathematics — Core Pure 1 · Chapter 8: Proof by Induction

Chapter 8: Proof by Induction

Why this matters

Proof by induction is the signature proof technique of Further Maths, and it ties together almost everything you've learned so far: the summation formulae from Chapter 3, and matrix multiplication from Chapter 6, all reappear here as things to prove. Beyond the exam, induction is one of the most important proof methods in all of mathematics — mastering the four-step structure here will serve you well beyond A Level.

Concept: the structure of an inductive proof

Proof by induction shows a statement is true for all positive integers n (or $n \geq$ some starting value) using four standard steps: **Basis** (show it's true for the starting value, usually $n = 1$), **Assumption** (assume it's true for some particular $n = k$), **Inductive step** (use that assumption to show it must then be true for $n = k + 1$), and **Conclusion** (state that, since it's true for $n = 1$ and each case implies the next, it's true for all n by the principle of mathematical induction). All four steps must be present and clearly signposted for full marks — examiners specifically credit the presence and wording of the conclusion.

It helps to picture an infinite line of dominoes. The basis step is knocking the first domino over. The inductive step is showing that *any* domino falling always knocks over the next one. Together, those two facts guarantee every single domino in the line eventually falls — even though you never touch domino number 1000 yourself. Proof by induction works the same way: you never verify the statement for every n directly, you just show the starting case holds and that truth always passes from one step to the next.

Key vocabulary

Term	Meaning
Basis case	The starting value of n (often $n = 1$) where the statement is directly verified
Inductive hypothesis	The assumption that the statement holds for $n = k$
Inductive step	Showing the statement for $n = k + 1$ follows from the $n = k$ assumption
Principle of mathematical induction	The logical rule that basis + inductive step together prove the statement for all $n \geq$ starting value

The basis case doesn't have to be $n = 1$ — a result might only hold from $n = 4$ onwards, say, in which case the basis case is $n = 4$ and the conclusion states the result for all $n \geq 4$. Always match the basis case to whatever starting value the question specifies.

Concept: induction for summation formulae

To prove a formula for $\sum_{r=1}^n f(r)$, the key algebraic move in the inductive step is always the same: take the assumed sum up to k , and add on the $(k+1)$ th term, then show the result matches the formula with n replaced by $k+1$.

Worked example 1: induction proof of a summation formula

EXAMPLE 1

Question. Prove by induction that $\sum_{r=1}^n r^2 = n(n+1)(2n+1)/6$ for all positive integers n .

Step 1 (Basis). When $n = 1$: LHS = $1^2 = 1$. RHS = $1(2)(3)/6 = 6/6 = 1$. Since LHS = RHS, the formula holds for $n = 1$. **(B1)**

Step 2 (Assumption). Assume the formula is true for $n = k$, i.e. $\sum_{r=1}^k r^2 = k(k+1)(2k+1)/6$. **(M1)**

Step 3 (Inductive step). Consider $n = k + 1$: $\sum_{r=1}^{k+1} r^2 = \sum_{r=1}^k r^2 + (k+1)^2 = k(k+1)(2k+1)/6 + (k+1)^2$, using the assumption from Step 2. **(M1)**

Step 4. Factor out $(k+1)/6$: $= [(k+1)/6] \times [k(2k+1) + 6(k+1)] = [(k+1)/6] \times [2k^2 + 7k + 6]$. **(M1 A1)**

Step 5. Factorise the quadratic: $2k^2 + 7k + 6 = (2k+3)(k+2)$, so the expression becomes $(k+1)(k+2)(2k+3)/6$, which is exactly the original formula with n replaced by $(k+1)$. **(A1)**

Step 6 (Conclusion). Since the formula holds for $n = 1$, and truth for $n = k$ implies truth for $n = k+1$, by the principle of mathematical induction the formula holds for all positive integers n . **(B1)**

Answer: The formula $\sum_{r=1}^n r^2 = n(n+1)(2n+1)/6$ is proven true for all positive integers n by induction.

Exam tip

EXAM TIP

The conclusion sentence is worth its own dedicated mark and is exactly what is most often skipped or garbled under time pressure. Memorise a fixed template sentence ("Since the statement is true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, by the principle of mathematical induction the statement is true for all [positive integers / $n \geq \dots$]) and write it out, word for word, at the end of every induction question — no exceptions.

Concept: induction for divisibility results

To prove a divisibility statement (e.g. " $f(n)$ is divisible by m for all n "), the inductive step has one recurring trick: **write $f(k+1)$ in terms of $f(k)$ plus (or minus) a multiple of m** , using the inductive hypothesis that $f(k) = m \times (\text{some integer})$.

Worked example 2: induction proof of a divisibility result

EXAMPLE 2

Question. Prove by induction that $5^n - 1$ is divisible by 4, for all positive integers n .

Step 1 (Basis). When $n = 1$: $5^1 - 1 = 4 = 4 \times 1$, which is divisible by 4. **(B1)**

Step 2 (Assumption). Assume $5^k - 1$ is divisible by 4 for $n = k$, i.e. $5^k - 1 = 4M$ for some integer

M. (M1)

Step 3 (Inductive step). Consider $n = k+1$: $5^{k+1} - 1 = 5 \times 5^k - 1 = 5(4M + 1) - 1$, using $5^k = 4M + 1$ from the assumption. (M1)

Step 4. Expand: $= 20M + 5 - 1 = 20M + 4 = 4(5M + 1)$. (M1 A1)

Step 5. Since $5M + 1$ is an integer, $5^{k+1} - 1$ is a multiple of 4, so the result holds for $n = k+1$. (A1)

Step 6 (Conclusion). Since the statement is true for $n = 1$, and true for $n = k$ implies true for $n = k+1$, by the principle of mathematical induction, $5^n - 1$ is divisible by 4 for all positive integers n . (B1)

Answer: Proven by induction as shown.

Exam tip

EXAM TIP

The critical algebraic trick in divisibility inductions is rewriting the $n = k+1$ expression to contain the $n = k$ expression as a visible sub-term (here, 5^k appears inside $5^{k+1} = 5 \times 5^k$). Practise spotting this "hidden" earlier term before attempting the substitution — it's the one genuinely new skill this sub-topic requires.

Concept: induction for matrix powers

To prove a formula for M^n (a matrix raised to a power), the inductive step uses $M^{k+1} = M^k \times M$, substitutes the assumed formula for M^k , and carries out the resulting matrix multiplication.

Worked example 3: induction proof for a matrix power

EXAMPLE 3

Question. Given $M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, prove by induction that $M^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ for all positive integers n .

Step 1 (Basis). When $n = 1$: $M^1 = M = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$, which matches $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ with $n = 1$. (B1)

Step 2 (Assumption). Assume $M^k = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix}$ for $n = k$. (M1)

Step 3 (Inductive step). Consider $n = k+1$: $M^{k+1} = M^k \times M = \begin{pmatrix} 1 & k \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$. (M1)

Step 4. Carry out the multiplication: top-left $= (1 \times 1) + (k \times 0) = 1$; top-right $= (1 \times 1) + (k \times 1) = 1 + k$; bottom-left $= (0 \times 1) + (1 \times 0) = 0$; bottom-right $= (0 \times 1) + (1 \times 1) = 1$. So $M^{k+1} = \begin{pmatrix} 1 & k+1 \\ 0 & 1 \end{pmatrix}$. (M1 A1)

Step 5. This matches the formula $\begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ with n replaced by $k+1$, so the statement holds for $n = k+1$. (A1)

Step 6 (Conclusion). Since the formula holds for $n = 1$, and truth for $n = k$ implies truth for $n = k+1$, by the principle of mathematical induction, $M^n = \begin{pmatrix} 1 & n \\ 0 & 1 \end{pmatrix}$ for all positive integers n . (B1)

Answer: Proven by induction as shown.

Exam tip

EXAM TIP

It is tempting to "prove" a matrix power formula by simply computing M^2 and M^3 and spotting a pattern — that's valuable for guessing the formula, but it is NOT a proof, and examiners will not award induction marks for pattern-spotting alone. Spotting the pattern is a legitimate first step (often given in the question), but the induction itself must follow the full four-step structure.

Chapter summary

Idea	Key point
Four-step structure	Basis, Assumption, Inductive step, Conclusion — all four required
Summation proofs	Add the $(k+1)$ th term to the assumed sum up to k , then factorise to match $n = k+1$
Divisibility proofs	Rewrite $f(k+1)$ to contain $f(k)$ as a visible sub-term, then substitute the assumption
Matrix power proofs	Use $M^{k+1} = M^k \times M$, substitute the assumed matrix, and multiply out
Conclusion sentence	A fixed, memorised template earns its own mark — never omit it

Common exam mistakes

- Omitting or garbling the final conclusion sentence, losing an easy mark.
- Assuming the result to prove it (circular reasoning) rather than genuinely deriving $n = k+1$ from $n = k$.
- Arithmetic or factorisation slips when combining the assumed sum with the extra term.
- For matrix inductions, multiplying $M \times M^k$ instead of $M^k \times M$ (usually harmless for these matrices, but sloppy practice that causes errors elsewhere).
- Believing that spotting a pattern from a few small cases constitutes a full proof.

You should now be able to

- State and correctly apply the four-step structure of proof by induction.
- Prove summation formulae by induction.
- Prove divisibility results by induction.
- Prove formulae for powers of a matrix by induction.
- Write a complete, correctly worded conclusion for an inductive proof.